

Fraud Detection System

Flow:

Data → Preprocessing → Feature Engineering → Scaling → Model Prediction → API → Dashboard
→ Explanation

Load transaction dataset with relevant features.

Perform preprocessing and clean the data.

Apply feature engineering (log transform, time features).

Select key features for model input.

Normalize features using StandardScaler.

Train Isolation Forest for anomaly detection.

Model outputs -1 (fraud) and 1 (normal).

Save model and scaler using joblib.

Build Flask API for real-time prediction.

Receive input in JSON format.

Apply preprocessing inside API.

Generate prediction using model.

Return result as JSON response.

Build Streamlit dashboard for visualization.

Add filters and search functionality.

Display transaction details and explanations.

Use GenAI for explanation generation.

Test system with sample data.

Visualize anomaly score distribution.

Combine all components into end-to-end system.